

Supplementary Information

Best of both worlds: Upskilling professionals in data science through online coursework and in-person codeathons

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Table S1: Learner survey responses for self-assessed comfort and confidence with data science, on a 1-5 scale. SD = standard deviation.

Measure	Timepoint	Mean	Median	SD
Comfort	Pre-course	2.66	2.00	0.987
Comfort	Post-course	3.96	4.00	0.759
Comfort	Post-codeathon	4.23	4.00	0.765
Confidence	Pre-course	2.30	2.00	1.090
Confidence	Post-course	3.89	4.00	0.641
Confidence	Post-codeathon	4.15	4.00	0.675

Table S2: Learner survey responses for self-assessed comfort and confidence with data science, paired two-sample Wilcoxon test statistics.

Measure	Comparison	Statistic	P-value	Adj. P-value
Comfort	Pre-course vs. Post-course	198	<0.001	<0.001
Comfort	Post-course vs. Post-codeathon	24.5	0.035	0.142
Confidence	Pre-course vs. Post-course	253	<0.001	<0.001
Confidence	Post-course vs. Post-codeathon	18	0.060	0.240

Table S3: Learner survey responses for self-assessed skills improvement with data science tools. SD = standard deviation.

Timepoint	Mean	Median	SD
Post-course	4.48	4.00	0.509
Post-codeathon	4.54	5.00	0.508

Table S4: Fixed Effects from Linear Mixed Model: Data Cleaning and Preprocessing. SE = standard error. DF = degrees of freedom.

Timepoint	Estimate	SE	DF	T Statistic	P-value	Adj. P-value
Intercept	0.208	0.026	72	8.05	<0.001	<0.001
Post-course	0.133	0.034	45	3.88	<0.001	0.003
Post-codeathon	0.178	0.035	51	5.13	<0.001	<0.001

Table S5: Fixed Effects from Linear Mixed Model: Reproducibility and Workflow Practices. SE = standard error. DF = degrees of freedom.

Timepoint	Estimate	SE	DF	T Statistic	P-value	Adj. P-value
Intercept	0.223	0.027	76	8.21	<0.001	<0.001
Post-course	0.188	0.038	49	4.96	<0.001	<0.001
Post-codeathon	0.279	0.041	53	6.77	<0.001	<0.001

Table S6: Fixed Effects from Linear Mixed Model: Analytical Reasoning and Data Exploration. SE = standard error. DF = degrees of freedom.

Timepoint	Estimate	SE	DF	T Statistic	P-value	Adj. P-value
Intercept	0.235	0.026	73	8.98	<0.001	<0.001
Post-course	0.182	0.033	49	5.57	<0.001	<0.001
Post-codeathon	0.246	0.034	55	7.14	<0.001	<0.001